

Bluetooth LE Channel Sounding

PBR Timing, Zephyr CS Event Reports, GATT Ranging, and an nRF54L15 Research Implementation

Research report

Prepared for a two-device nRF54L15 ranging platform

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Scope

This report explains the Bluetooth Core Channel Sounding mechanisms, the adopted Ranging Service (RAS) and Ranging Profile (RAP), their role relationships, and a standards-aligned implementation and evaluation path using two nRF54L15 devices.

Executive summary

Bluetooth LE Channel Sounding (CS), introduced in Bluetooth Core 6.0, is a controller and Link Layer feature for acquiring phase and timing observations over a dedicated set of 1 MHz channels. It supports phase-based ranging (PBR) for fine distance estimation and round-trip timing (RTT) for an independent estimate and stronger resistance to relay attacks. The Core Specification deliberately does not standardize the final distance-estimation algorithm; it standardizes the radio exchanges, control procedures, and measurement reporting. [1], [2]

RAS and RAP solve a different problem. A CS procedure produces local results at both radios. The Ranging Service is a GATT service by which one device exposes its controller-generated ranging data. The Ranging Profile specifies how a Ranging Requester discovers that service, selects real-time or on-demand delivery, reassembles segmented data, and passes the complete remote result to a distance-measurement application. The application combines local and remote data and computes distance. RAS/RAP transport results; they neither perform the Channel Sounding radio procedure nor define a ranging algorithm. [4], [5]

Central conclusion Channel Sounding is not itself a GATT profile. For an interoperable two-device product, use Core CS for measurement acquisition and the adopted RAS/RAP specifications for peer-result transfer. A private GATT characteristic can work for experiments, but it is not the standard interoperability path.

Key findings

- Four role systems coexist and must not be conflated: GAP Central/Peripheral, GATT Client/Server, CS Initiator/Reflector, and RAP Ranging Requester/Responder.
- The common nRF54L15 mapping is Initiator = Central = GATT client = Ranging Requester, and Reflector = Peripheral = GATT server = Ranging Responder; the Core and profile specifications allow other mappings.
- Mode 0 calibrates frequency offset; mode 1 supports RTT; mode 2 supports PBR; optional mode 3 combines RTT and PBR. A procedure begins with one or more mode-0 steps.
- Zephyr delivers each CS result fragment through `le_cs_subevent_data_available`. The Host must preserve the report header, append the length-delimited step records, and use the subevent/procedure status fields rather than assuming one callback equals one procedure.
- RAS requires encryption for its characteristics and supports mandatory on-demand transfer plus optional real-time transfer, including segmentation, acknowledgements, overwrite reporting, and optional recovery of lost segments.
- For two nRF54L15 DKs, Nordic's RAS Initiator and RAS Reflector samples are the best standards-aligned baseline. Zephyr's `connected_cs` and `cs_test` samples remain useful for protocol learning and controlled experiments, but their simple private GATT exchange is not equivalent to RAP/RAS.

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1. Scope, terminology, and standards status

1.1 Scope

The report covers connected, one-to-one Bluetooth LE Channel Sounding between two devices. It focuses on the Core procedure, Host Controller Interface (HCI) reporting, GATT-based peer result exchange, and application-level distance estimation. It does not attempt to reproduce the normative timing tables, conformance requirements, or radio test procedures in full; the adopted specifications remain authoritative.

1.2 Standards status as of 19 July 2026

Channel Sounding was introduced by Bluetooth Core 6.0. Bluetooth Core 6.3 is currently adopted and retains the baseline design while adding, among other items, optional inline phase-correction term transfer and PHY-specific RTT accuracy reporting. This report explains the interoperable Core 6.0 baseline and treats later enhancements as optional capabilities that must be discovered rather than assumed. [1], [3]

Ranging Service 1.0 and Ranging Profile 1.0 are adopted Bluetooth specifications. The RAS specification page identifies Erratum 26610 as mandatory when claiming RAS 1.0 compliance. Implementations and research reports should therefore identify the exact Core, RAS/RAP, SDK, controller, and errata versions used. [4], [5]

1.3 Terminology

Term	Meaning in this report
Channel Sounding (CS)	Controller/Link Layer procedures and RF exchanges used to obtain timing and phase observations.
PBR	Phase-based ranging; estimates distance from phase evolution across multiple frequencies.
RTT	Round-trip timing; estimates distance from bidirectional propagation time after turnaround and circuit-delay effects are handled.
Ranging Data	Controller-generated CS subevent results delivered to the Host and, when remote transfer is required, conveyed by RAS.
Distance estimate	Application output computed from local and peer Ranging Data; not a direct Core controller output.
Initiator / Reflector	CS radio roles. The Initiator performs the exchanges for distance estimation; the Reflector returns the specified packets or tones.
RREQ / RRSP	RAP roles: Ranging Requester and Ranging Responder.

2. Protocol architecture and role separation

2.1 Where Channel Sounding sits

CS adds a dedicated LE Channel Sounding physical channel and physical link. It affects the Physical Layer, Link Layer, HCI, and GAP-related control and security definitions. It does not create an application data bearer: the Bluetooth SIG technical overview explicitly notes that no logical transport or logical link is associated with the CS physical link. Configuration and measurement reporting pass between Host and Controller through HCI. [2]

Layer	Responsibility in a ranging system
Application	Selects operating parameters, combines local and peer observations, estimates distance, filters results, and applies use-case policy.
RAP / RAS over GATT	Discovers, configures, segments, transfers, acknowledges, and optionally retransmits remote Ranging Data.
HCI	Carries CS commands to the controller and LE CS Subevent Result / Continue events to the Host.
Link Layer	Performs CS security, capabilities, configuration, start, channel selection, scheduling, and CS step exchanges.
Physical Layer	Generates and receives CS_SYNC packets and CS tones, switches frequency and antenna paths, and obtains timing/phase measurements.

2.2 Four independent role axes

The same device participates in several protocol layers, and each layer assigns its own role. The role names answer different questions and are not synonyms. In particular, a CS Initiator need not be a GAP Central, and a CS Reflector need not be a GATT server. RAP imposes a GATT client/server relationship for its own exchange, while the Core permits either Central or Peripheral to assume either CS role. [2], [4], [5]

Role system	Role A	Role B	What it controls
GAP / LE connection	Central	Peripheral	Connection establishment and ACL scheduling.
GATT	Client	Server	Who discovers/requests attributes and who exposes them.
Core Channel Sounding	Initiator	Reflector	Who initiates CS tones/packets and who returns the specified response.
Ranging Profile	RREQ	RRSP	Who requests remote Ranging Data and who makes it available through RAS.

2.3 Common two-nRF54L15 role mapping

Device	GAP	GATT	CS	RAP	Application responsibility
Device A	Central	Client	Initiator	RREQ	Stores local results, fetches peer data, computes and logs distance.
Device B	Peripheral	Server	Reflector	Exposes RAS, stores controller results, and transfers them on request or in real time.	

Terminology caution The Bluetooth SIG profile uses Ranging Requester and Ranging Responder. Nordic sample names have also used the spelling Requestor. These RAP roles are not the same as the CS Initiator and Reflector roles, even though the Nordic reference applications pair them in the common way.

3. Distance-measurement principles

3.1 Phase-based ranging

PBR uses two-way tone exchange at multiple RF frequencies. At each selected frequency, the Initiator and Reflector exchange CS tones and obtain phase correction terms; combining both devices' observations forms a round-trip phase measurement. Propagation introduces a phase rotation proportional to frequency and distance. Taking phase differences across frequencies removes a constant phase offset and turns the phase slope into a distance observation. [2]

$$d \approx (c / 4\pi) \cdot |\Delta\phi / \Delta f|$$

Ideal two-way, unwrapped phase model; d is one-way distance, c is the speed of light, $\Delta\phi$ is two-way phase change, and Δf is frequency separation.

Practical estimators use many channels rather than only two. A phase-slope estimator unwraps phase and fits a line versus frequency; an inverse fast Fourier transform (IFFT) estimator maps the frequency samples to a delay-like profile and selects a peak. Multipath produces several components, incomplete channel sets create sidelobes, and antenna/circuit phase varies with frequency. Consequently, calibration, robust estimation, channel-quality filtering, and temporal tracking often dominate real performance. The Core Specification standardizes the observations, not the estimator. [2], [7]

3.2 Round-trip timing

RTT uses CS_SYNC packet exchanges. The Initiator measures elapsed time between its transmission and reception of the response; the Reflector's turnaround contribution is removed using the corresponding timing information. After correction, half of the bidirectional propagation time multiplied by the speed of light gives one-way range. [2]

$$d \approx (c / 2) \cdot [(ToA_i - ToD_i) - (ToD_r - ToA_r)]$$

Conceptual RTT equation. Real implementations also account for controller-reported accuracy, quantization, and calibrated circuit delays.

Timing resolution at Bluetooth bandwidths generally makes RTT less precise than PBR for fine ranging. Its main value is diversity: it responds differently to phase ambiguity, multipath, and some attacks. The Bluetooth SIG recommends combining PBR and RTT so that the estimates can be cross-checked. [2]

3.3 CS step modes

Mode	Purpose	Exchange	Support
0	Frequency-offset calibration	CS_SYNC plus reflector tone; Initiator derives compensation information.	Mandatory
1	RTT	CS_SYNC packet in each direction with controlled timing.	Mandatory
2	PBR	CS tones in both directions across one or more antenna paths.	Mandatory
3	Combined RTT and PBR	CS_SYNC packets and CS tones in one step.	Optional

Each CS subevent begins with one or more mode-0 steps. The application then selects a main mode and, optionally, a submode. If mode 3 is not mutually supported, a common secure fine-ranging configuration uses mode 2 as the main mode and periodically inserts mode-1 steps. More channels and more step repetitions usually provide more information, at the cost of airtime, latency, energy, and sensitivity to motion during the measurement window. [2]

3.4 Mode-2 PBR step timing

A mode-2 step contains a directional tone block from the Initiator, ramp-down, an interlude, and the reciprocal directional tone block from the Reflector. Each block reserves N_{AP} antenna-path tone slots plus one extension slot. The receiving side measures one Phase Correction Term (PCT) per occupied path position and for the extension position. Figure 1 shows the exact timing relationship and the one-antenna example developed in the PBR timing analysis. [1], [2]

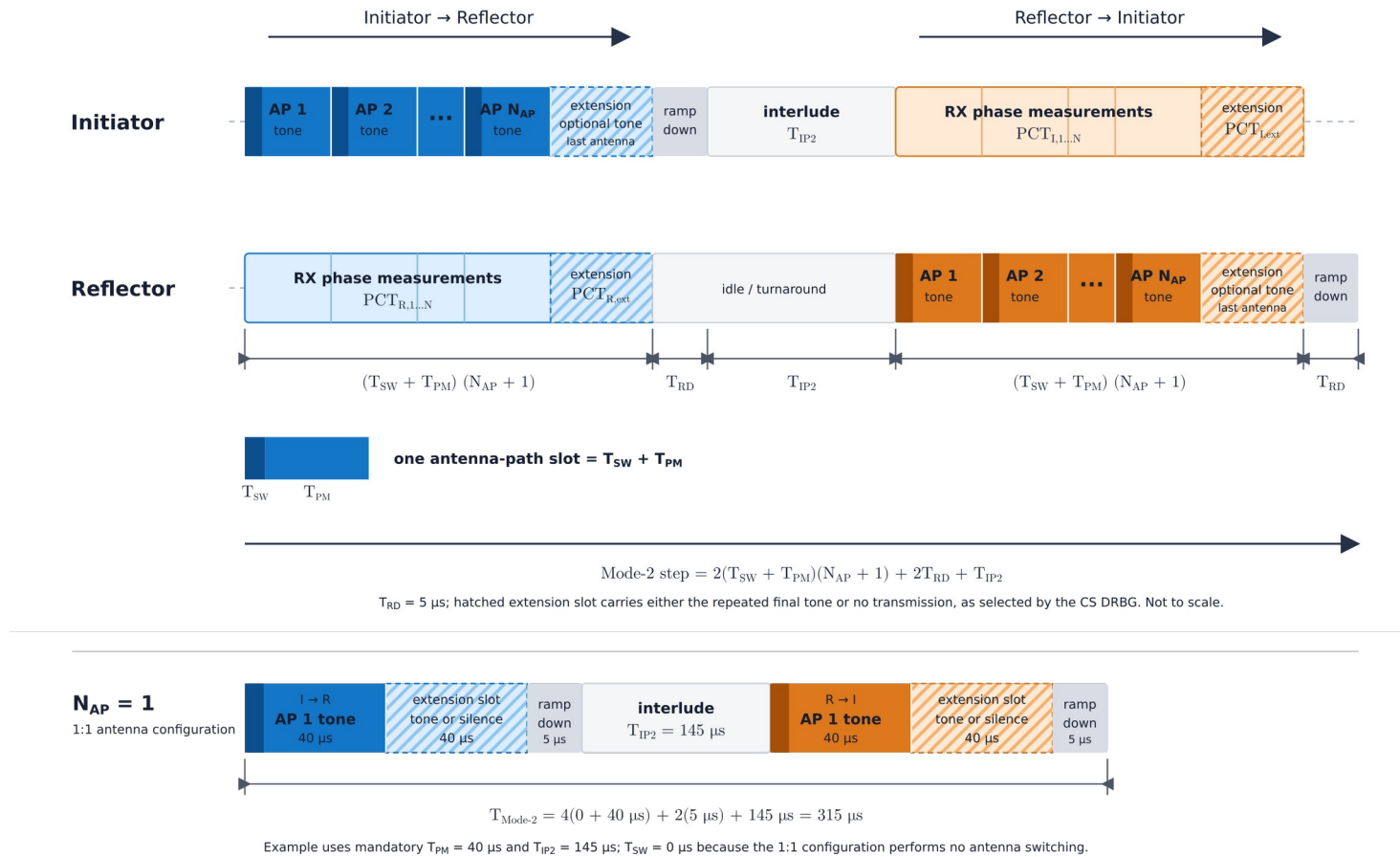


Figure 1. Mode-2 PBR step timing, including the $N_{AP} = 1$ numerical example. Not to scale.

3.4.1 Symbols and duration calculation

Symbol	Meaning
AP1 ... AP N_AP	Antenna-path positions used for the phase-measurement tones. The permutation index determines which physical antenna path occupies each tone position.
N_AP	Number of antenna paths used during the phase-measurement stage (1 to 4). Each directional tone block has N_AP + 1 timed positions because the extension slot is included.
T_SW	Antenna switching interval associated with each path position. In the 1:1 example there is no switching, so T_SW = 0 μs.
T_PM	Phase-measurement tone interval for one path position or the extension position. The example uses 40 μs.
T_RD	Transmitter ramp-down interval after a directional tone block. The example uses 5 μs.
T_IP2	Interlude between the Initiator-to-Reflector and Reflector-to-Initiator CS-tone blocks. The example uses 145 μs.
PCT_R,k / PCT_I,k	Complex Phase Correction Term measured by the Reflector (R) or Initiator (I) for tone position k. Zephyr carries each HCI-formatted PCT in three octets.
Extension slot	The extra timed position after the N_AP path tones. It contains a repeated tone or silence according to the procedure; its time remains reserved in either case.
T_Mode-2	Total duration of the mode-2 step, excluding the frequency-change interval that separates adjacent steps.

$$T_{\text{Mode-2}} = 2(T_{\text{SW}} + T_{\text{PM}})(N_{\text{AP}} + 1) + 2T_{\text{RD}} + T_{\text{IP2}}$$

The factor of two accounts for the two directional tone blocks. The (N_AP + 1) factor includes the extension position.

For the one-antenna configuration shown in Figure 1, N_AP = 1, T_SW = 0 μs, T_PM = 40 μs, T_RD = 5 μs, and T_IP2 = 145 μs. Therefore T_Mode-2 = 2(0 + 40)(1 + 1) + 2(5) + 145 = 315 μs. The RF exchange contains AP1 plus one extension position in each direction; the extension position still consumes 40 μs when it is silent. [1], [2]

4. Channel Sounding procedure lifecycle

4.1 Connection and preparation

CS operates between two connected LE devices. The normal sequence establishes an LE ACL connection and then prepares the controllers. Some information may be cached, so every control procedure need not occur on every connection. [2]

- 1. Create the LE ACL connection.** A Central connects to a Peripheral. CS radio roles are assigned separately from these GAP roles.
- 2. Establish CS security.** The controllers exchange CS-specific random material used by the CS deterministic random-bit generator and security mechanisms. This is distinct from, but initialized over, the ACL connection.
- 3. Exchange capabilities.** The peers determine mutual PHYs, modes, antenna-path support, RTT accuracy, attack-detection support, timing limits, and related features.
- 4. Create a CS configuration.** A mutually supported configuration identifies the Initiator and Reflector, channel map, step-mode arrangement, PHY, antenna configuration, and timing choices.
- 5. Obtain optional calibration data.** The Initiator can request and cache the Reflector's mode-0 frequency actuation error table where applicable.

- 6. Set procedure parameters and enable.** The Host selects repetition, event/subevent scheduling, step counts, and other parameters; the controllers agree on a start point anchored to the ACL connection.

4.2 Procedure, event, subevent, and step

A CS procedure contains CS events; an event contains one or more subevents; a subevent contains steps. Packets and tones are exchanged inside steps. Frequency-hop intervals separate steps, and each subevent begins with calibration. The hierarchy allows an application to trade update rate, observation diversity, and resource consumption. [2]

Unit	Function	Research-relevant controls
Procedure	One logical measurement run, optionally repeated.	Repeat count, interval, termination policy.
Event	Groups subevents and anchors them relative to ACL events.	Event interval and scheduling coexistence.
Subevent	Groups a coherent set of steps and produces controller result events.	Subevent interval, duration, number of subevents.
Step	Performs calibration, RTT, PBR, or combined exchange on a selected channel.	Mode, channel, PHY, antenna path, timing, quality filtering.

4.3 Controller-to-Host result reporting

The controller reports observations using LE CS Subevent Result and LE CS Subevent Result Continue HCI events. Zephyr normalizes those controller events for connected CS into `bt_conn_le_cs_subevent_result` and invokes `bt_conn_cb.le_cs_subevent_data_available`. The callback receives the connection separately, a decoded report header, and a `net_buf_simple` containing zero or more encoded step records. The application should copy or consume the buffer during the callback; it must not retain the pointer as durable storage. [11], [12], [13]

```
struct bt_conn_le_cs_subevent_result {
    struct {
        uint8_t config_id;
        uint16_t start_acl_conn_event;
        uint16_t procedure_counter;
        uint16_t frequency_compensation;
        int8_t reference_power_level;
        enum bt_conn_le_cs_procedure_done_status procedure_done_status;
        enum bt_conn_le_cs_subevent_done_status subevent_done_status;
        enum bt_conn_le_cs_procedure_abort_reason procedure_abort_reason;
        enum bt_conn_le_cs_subevent_abort_reason subevent_abort_reason;
        uint8_t num_antenna_paths;
        uint8_t num_steps_reported;
        uint8_t abort_step;
    } header;
    struct net_buf_simple *step_data_buf;
};
```

4.3.1 Raw Zephyr HCI event layouts

At the HCI boundary, the first report and its continuation are different packed event layouts. The first event carries the procedure identity and calibration metadata; Continue deliberately omits those repeated fields and carries the remaining step bytes. Zephyr's Host translates both forms into the connected-CS callback described above. [13]

```
struct bt_hci_evt_le_cs_subevent_result {
    uint16_t conn_handle;
    uint8_t  config_id;
    uint16_t start_acl_conn_event_counter;
    uint16_t procedure_counter;
    uint16_t frequency_compensation;
    uint8_t  reference_power_level;
    uint8_t  procedure_done_status;
    uint8_t  subevent_done_status;
    uint8_t  subevent_abort_reason : 4;
    uint8_t  procedure_abort_reason : 4;
    uint8_t  num_antenna_paths;
    uint8_t  num_steps_reported;
    uint8_t  steps[];
};

struct bt_hci_evt_le_cs_subevent_result_continue {
    uint16_t conn_handle;
    uint8_t  config_id;
    uint8_t  procedure_done_status;
    uint8_t  subevent_done_status;
    uint8_t  subevent_abort_reason : 4;
    uint8_t  procedure_abort_reason : 4;
    uint8_t  num_antenna_paths;
    uint8_t  num_steps_reported;
    uint8_t  steps[];
};
```

4.4 Zephyr report-header fields

Field	Type / units	Meaning and handling
config_id	uint8_t; 0–3	CS configuration that produced the result. It is zero and has no meaning for CS Test output.
start_acl_conn_event	uint16_t counter	ACL connection-event counter at which the reported CS procedure started. It is zero and has no meaning for CS Test output.
procedure_counter	uint16_t counter	Procedure count since completion of the CS Security Start procedure. Use it to group fragments and subevents from the same procedure; RAS derives its ranging counter from this value.
frequency_compensation	uint16_t storage; signed 15-bit value, 0.01 ppm	Initiator frequency-compensation observation, nominal range –100 to +100 ppm. Test the Zephyr unavailable sentinel before sign extension or arithmetic.
reference_power_level	int8_t dBm; –127 to +20	Reference power observed during the subevent. Test BT_HCI_LE_CS_REF_POWER_LEVEL_U NAVAILABLE before use.
procedure_done_status	enum	COMPLETE on the final report of a successful procedure; INCOMPLETE while additional subevents/results belong to it; ABORTED when the procedure terminated abnormally.
subevent_done_status	enum	COMPLETE or ABORTED. On abort, consult subevent_abort_reason and abort_step.

Field	Type / units	Meaning and handling
procedure_abort_reason	enum	Meaningful when procedure_done_status is ABORTED; otherwise it indicates not aborted.
subevent_abort_reason	enum	Meaningful when subevent_done_status is ABORTED; Zephyr defines requested, missing CS_SYNC, scheduling conflict, and unspecified reasons.
num_antenna_paths	uint8_t	N_AP used during the phase-measurement stage. It determines how many tone_info records are expected in mode 2/3 step data.
num_steps_reported	uint8_t	Number of length-delimited CS step records in this callback/report fragment; it is not necessarily the total number in the full procedure.
abort_step	uint8_t step index	First aborted step. It is 255 when the subevent completed; if abort_step = 100, steps 0–99 are complete and step 100 onward is aborted.
step_data_buf	net_buf_simple *	Encoded step records; NULL when num_steps_reported is zero. bt_le_cs_step_data_parse() consumes this buffer while calling the parser callback.

4.5 Step record envelope and mode-specific payloads

bt_le_cs_step_data_parse() decodes each record into bt_le_cs_subevent_step { mode, channel, data_len, data }. The mode byte selects the payload layout and data_len bounds it; therefore the parser must validate the expected minimum length before casting data. The helper consumes the net_buf_simple. Applications that also need the raw bytes should clone or copy them first. [12]

```

struct bt_le_cs_subevent_step {
    uint8_t mode;
    uint8_t channel;
    uint8_t data_len;
    const uint8_t *data;
};

struct bt_hci_le_cs_step_data_tone_info {
    uint8_t phase_correction_term[3];
    uint8_t extension_indicator : 4;
    uint8_t quality_indicator : 4;
};

```

Step mode	Zephyr payload content	Measurement use
Mode 0	Initiator: packet quality (AA check and bit errors), RSSI, packet antenna, measured frequency offset. Reflector has the corresponding packet-quality/RSSI/antenna fields.	Calibrate or qualify frequency offset and CS_SYNC reception before ranging steps.
Mode 1	Packet quality, NADM, RSSI, Initiator ToA–ToD or Reflector ToD–ToA timing, and packet antenna. SS-RTT variants also include packet PCT fields.	RTT observation and attack/quality screening.
Mode 2	Antenna permutation index followed by tone_info[N_AP + 1]. Each tone_info holds phase_correction_term[3], a 4-bit extension indicator, and a 4-bit quality indicator.	PBR. Parse each 24-bit little-endian PCT with bt_le_cs_parse_pct(); map tone position to physical antenna path with bt_le_cs_get_antenna_path().
Mode 3	Mode-1 packet/timing fields plus antenna permutation index and tone_info[N_AP + 1]; SS-RTT variant adds packet PCT fields.	Combined RTT and PBR observations from one step.

PCT representation Zephyr's HCI tone record stores the PCT in three octets: two signed 12-bit components (I and Q) packed little-endian. `bt_le_cs_parse_pct()` returns sign-extended `int16_t` values. Do not interpret the three bytes as one scalar phase angle; compute phase from $\text{atan2}(Q, I)$ after quality gating.

4.6 Mapping report type and status to procedure timing

A report is tied to a CS subevent, whereas `procedure_done_status` describes the enclosing CS procedure. The first HCI LE CS Subevent Result for a subevent carries the complete metadata and the first group of step bytes. If the step set does not fit, the controller sends one or more LE CS Subevent Result Continue events with a reduced header and the remaining step bytes. Zephyr presents both report types through the same application callback. [1], [11], [13]

Point in CS timeline	Controller report	Status interpretation / Host action
After a subevent has produced reportable steps	LE CS Subevent Result	Full header: connection/configuration, procedure identity, ACL anchor, frequency compensation, reference power, statuses, <code>N_AP</code> , and first steps. Start or locate the accumulator keyed by connection + <code>config_id</code> + <code>procedure_counter</code> .
The same subevent has more result bytes than fit in the first event	One or more LE CS Subevent Result Continue events	Reduced HCI header plus remaining steps. Append in arrival order to the same accumulator. At the Zephyr callback layer, process the normalized header/status and <code>step_data_buf</code> identically.
A subevent completes but the procedure has more scheduled subevents	Result/Continue with subevent COMPLETE and procedure INCOMPLETE	Finalize the subevent boundary if the application stores subevents separately, but retain the procedure accumulator.
Final successful subevent and final fragment	Result/Continue with procedure COMPLETE	The complete set may now be handed to the estimator or RAS database. Do not finalize earlier merely because a subevent is complete.
RF or scheduling failure	Result/Continue with subevent and/or procedure ABORTED	Keep only steps before <code>abort_step</code> if useful for diagnostics; record both abort reasons and normally reject the procedure for distance estimation.

Figure 2. Logical mapping of CS RF execution to controller result-report timing. Report delivery follows controller buffering and HCI transport, so it is not an RF time-of-flight measurement.

```

CS procedure P(k)
  CS event 0
    subevent 0: M0 ... M2/M1/M3 steps -> Result -> Continue*
                  subevent=COMPLETE, procedure=INCOMPLETE
    subevent 1: M0 ... M2/M1/M3 steps -> Result -> Continue*
                  subevent=COMPLETE, procedure=INCOMPLETE
    ...
  final subevent: M0 ... ranging steps -> Result -> Continue*
                  subevent=COMPLETE, procedure=COMPLETE
  * zero or more continuation reports; abort status may terminate either scope

```

Important data-flow fact Both controllers generate local CS results. If the distance algorithm runs only on Device A, Device A already has its own HCI results but still needs Device B's results. RAS/RAP standardize that peer-result transfer.

5. Ranging Service and Ranging Profile

5.1 What RAP and RAS standardize

RAP defines a Ranging Requester (RREQ), which is a GATT client attached to a distance-measurement application, and a Ranging Responder (RRSP), which is a GATT server implementing RAS. The RRSP makes controller-generated Ranging Data available; the RREQ configures delivery and receives the data. RAP allows a device to implement both profile roles and does not force a particular Central/Peripheral topology. [5]

RAS defines the GATT service and data procedures. The server-side controller may itself be a CS Initiator, Reflector, or both. Thus RAS does not equate GATT Server with CS Reflector. The common Nordic sample mapping is a design choice that makes the architecture intuitive, not a universal protocol law. [4]

5.2 RAS characteristic set

Characteristic	Status	Principal use	Required security
RAS Features	Mandatory	Read server support for real-time transfer, lost-segment recovery, abort, and filtering.	Encryption
Real-time Ranging Data	Optional	Notify or indicate data as subevent/procedure results arrive from the controller.	Encryption
On-demand Ranging Data	Mandatory	Notify or indicate stored data after a control-point request.	Encryption
RAS Control Point	Mandatory	Get/acknowledge data, optionally recover segments, set filters, or abort.	Encryption
Ranging Data Ready	Mandatory	Announce the counter of data available for on-demand retrieval.	Encryption
Ranging Data Overwritten	Mandatory	Report that unread on-demand data was replaced by a newer result.	Encryption

5.3 Segmentation and reassembly

A full procedure result usually exceeds one ATT message. RAS prepends a one-octet segmentation header carrying First Segment, Last Segment, and a six-bit rolling segment index. Payload per message is limited to ATT_MTU minus four octets. The receiver concatenates segments in order to recover the Ranging Data Body. With multiple EATT bearers, arrival order is not guaranteed and the profile directs the client to reorder by segment index. [4], [5]

The Ranging Data Body contains a Ranging Header followed by one or more subevent headers and step data. Filtering can reduce transfer and processing load by omitting fields not required by the selected estimator, but filtering must preserve every observation and quality indicator needed for reproducible research. [4]

5.4 Real-time delivery

If supported, the RREQ enables notifications or indications on Real-time Ranging Data. The RRSP can begin transfer after complete subevent results arrive from its controller, potentially while the overall CS procedure is still running. This minimizes latency and storage but does not provide the same recovery semantics as the on-demand path; unread data may be replaced by newer data. [4], [5]

5.5 On-demand delivery

1. **Subscribe.** The RREQ enables On-demand Ranging Data, Ranging Data Ready, Ranging Data Overwritten, and RAS Control Point indications or notifications as required.
2. **Receive ready indication.** The RRSP announces a Ranging Counter after data becomes available.
3. **Request the record.** The RREQ writes Get_Ranging_Data with the counter to the RAS Control Point.
4. **Transfer and reassemble.** The RRSP sends segmented On-demand Ranging Data, followed by a complete response at the control point.
5. **Recover if necessary.** If supported, the RREQ requests missing segment indices; otherwise it treats the record as incomplete.
6. **Acknowledge.** After successful reconstruction, the RREQ writes ACK_Ranging_Data; the RRSP may then delete the stored record.

The server must retain at least one complete procedure result, but a new result can overwrite an unread record when memory is constrained. A research implementation should log the Ranging Counter, segment indices, overwrite events, and completion status so that missing-data bias is visible in the dataset. [4]

5.6 What RAS/RAP do not standardize

- They do not schedule or execute the RF Channel Sounding procedure; Core controller commands and Link Layer procedures do that.
- They do not produce a final distance characteristic. Ranging Data consists of procedure results used by an application algorithm.
- They do not define phase unwrapping, IFFT processing, multipath mitigation, calibration, fusion, tracking, outlier rejection, or application thresholds.
- They do not make every pair of implementations equally accurate; interoperability of measurements and transport does not imply identical estimator performance.

6. End-to-end operation

6.1 Standards-aligned sequence

- 1. Discover and connect.** Device A scans for a Peripheral advertising the Ranging Service UUID, connects, performs service discovery, and establishes the required link security.
- 2. Configure the GATT path.** Device A reads RAS Features and selects real-time or on-demand delivery. It configures the required CCC descriptors and any optional filter.
- 3. Prepare Core CS.** The Hosts request capability exchange, enable CS security, create a compatible CS configuration, and set procedure scheduling parameters.
- 4. Execute CS steps.** The controllers exchange mode-0 calibration steps and the selected RTT/PBR steps across the configured channel set.
- 5. Collect local results.** Each controller reports subevent results to its own Host through HCI events. Device A stores its local result; Device B passes its result into the RAS server database.
- 6. Transfer peer results.** RAS sends Device B's Ranging Data to the RREQ on Device A using real-time or on-demand procedures.
- 7. Validate and pair records.** Device A matches local and peer records using procedure identifiers/counters, checks completeness and quality, and discards or marks invalid observations.
- 8. Estimate and report distance.** The distance-measurement application applies PBR and RTT estimators, security/quality checks, calibration, and temporal filtering before exposing a result.

6.2 Why both data sets are needed

Two-way PBR and RTT are designed to cancel or estimate quantities that cannot be inferred from one device's observation alone. The application therefore treats local and peer results as a paired measurement. Pairing the wrong subevents, silently losing segments, or mixing different configurations can create plausible but invalid distances. Dataset schemas should retain the connection identity, CS configuration identifier, procedure and subevent status, Ranging Counter, step index, channel, mode, antenna path, and quality fields.

6.3 Private GATT exchange versus RAS/RAP

Aspect	Private characteristic used by simple samples	Adopted RAS/RAP
Purpose	Minimal demonstration or experiment-specific data exchange.	Interoperable transfer of controller-generated Ranging Data.
Data model	Implementation-defined; often bounded by a fixed buffer.	Defined Ranging Data Body with procedure/subevent/step content.
Segmentation	Application-specific.	Standard first/last flags and rolling segment index.
Delivery	Usually one custom notification/write flow.	Mandatory on-demand and optional real-time modes.
Loss handling	Application-specific or absent.	Acknowledgement, overwrite reporting, and optional lost-segment retrieval.
Interoperability	Only with software that understands the private UUID and encoding.	Designed for conforming RAP/RAS peers.

7. nRF54L15 implementation strategy

7.1 Recommended baseline

For two nRF54L15 devices, use Nordic's Channel Sounding Initiator with Ranging Requester sample on Device A and Channel Sounding Reflector with Ranging Responder sample on Device B. The former runs as a BLE Central and GATT RAS client, starts regular CS procedures, stores local subevent data, fetches peer Ranging Data, and computes distance. The latter runs as a BLE Peripheral, advertises RAS, configures the CS Reflector role, and stores controller results for the RREQ. [8], [9]

Nordic documents qualified Channel Sounding host/controller support for the nRF54L Series in nRF Connect SDK from version 3.0.1 onward and identifies the nRF54L15 as a supported device. The exact sample directory layout has changed between SDK releases, so builds should use the documentation shipped with the installed SDK rather than copying a path from another release. [10]

Build target For the nRF54L15 DK, the current board target is typically `nrf54l15dk/nrf54l15/cpuapp`. Confirm it against the selected nRF Connect SDK release and the sample's generated build instructions.

7.2 When to use the other Channel Sounding samples

Sample family	Best use	GATT relationship	Research caution
RAS Initiator + RAS Reflector	Interoperable product baseline and standards-aligned research.	Uses the adopted Ranging Service/Profile flow.	Still uses a vendor/application ranging algorithm; record its version and parameters.
<code>connected_cs</code>	Learning normal connected CS setup and directly examining Core CS APIs.	Zephyr example exchanges results with a simple custom characteristic.	Do not describe the private characteristic as RAS/RAP.
<code>cs_test</code>	Controlled RF experiments and overriding randomized parameters.	Uses a simple sample-specific exchange.	Test behavior is valuable for characterization but is not normal production operation.

Zephyr's `connected_cs` and `cs_test` samples include a basic distance estimator and exchange both devices' results through a private GATT characteristic. The Zephyr documentation notes an approximately 512-byte sample limit, sufficient for a small number of RTT observations and PBR observations across roughly 35-40 channels with one antenna path. This is useful for learning, but it is a sample constraint, not a Bluetooth Channel Sounding limit. [7]

7.3 Suggested firmware decomposition

Module	Responsibilities
Connection manager	Advertising/scanning, connection lifecycle, pairing/bonding, parameter updates, reconnect policy.
CS manager	Capability discovery/cache, CS security, configuration creation, procedure parameters, enable/disable state machine.
Local result collector	Non-blocking aggregation of HCI result and continue events; completeness and error tracking.
RAS server/client	Feature negotiation, CCC handling, real-time/on-demand transfer, segmentation, reassembly, loss/overwrite handling.

Module	Responsibilities
Record synchronizer	Matches local and remote subevents and rejects inconsistent configuration, counters, or incomplete results.
Estimator	PBR phase/IFFT processing, RTT processing, cross-checks, quality gating, calibration, temporal filtering.
Research logger	Raw observations, metadata, firmware/configuration hashes, ground truth, and final estimator outputs.

7.4 Minimum configuration and log record

- Firmware, nRF Connect SDK, Zephyr, controller, Core, RAS/RAP, and mandatory erratum versions.
- Board revision, antenna, enclosure, supply state, transmit power, PHY, connection interval, and temperature if available.
- CS role, GAP role, RAP role, configuration identifier, channel map, step modes, step count, procedure/subevent intervals, and antenna configuration.
- All local and peer step data required to reproduce PBR and RTT estimates, plus packet/tone quality, RSSI, NADM, completion status, and discarded-channel mask.
- Estimator name/version, calibration constants, phase-unwrapping or IFFT parameters, outlier rules, smoothing state, and reported uncertainty.
- Ground-truth distance, orientation, motion state, environment class, timestamp, trial identifier, and operator notes.

8. Research methodology

8.1 Research questions

A reproducible study should separate protocol functionality from estimator performance. Example research questions are: (i) how accuracy and availability change with distance, orientation, and multipath; (ii) how PBR, RTT, and fused estimates differ; (iii) how channel count and procedure timing affect latency and energy; and (iv) how well quality and attack metrics predict large errors.

8.2 Experimental matrix

Factor	Suggested levels	Reason
Distance	Dense short-range points (e.g., 0.25-3 m) plus longer points up to the use-case limit.	Characterize near-field behavior, bias, ambiguity, and range-dependent variance.
Environment	Open LOS, furnished room, corridor, and controlled NLOS/obstruction.	Expose multipath and blockage effects.
Orientation	At least four relative rotations; repeat with intended enclosure/body placement.	Capture antenna gain, polarization, and group-delay changes.
Motion	Static, slow approach/retreat, lateral motion, and representative application motion.	Test coherence over multi-channel procedures and tracking lag.
Configuration	Channel subsets/count, procedure interval, main/submode ratio, PHY, TX power, antenna path.	Quantify accuracy-latency-energy trade-offs.
Estimator	PBR phase slope, PBR IFFT, RTT, fused estimate, and selected filters.	Separate standardized observations from algorithmic performance.

8.3 Measurement procedure

- 1. Fix the reference geometry.** Define antenna reference points, not enclosure edges, and measure ground truth with uncertainty substantially smaller than the expected ranging error.
- 2. Warm up and identify devices.** Record board identifiers and allow temperature-sensitive oscillators and electronics to reach the chosen operating condition.
- 3. Capture raw paired results.** Store local and RAS-transferred peer records before estimator-specific filtering. Do not retain only the printed distance.
- 4. Randomize trial order.** Avoid confounding distance or orientation with time, temperature drift, battery state, or changing RF occupancy.
- 5. Repeat independently.** Use enough repeated procedures per condition and repeat the full setup on different days or device pairs when generalization matters.
- 6. Report failures.** Treat missing, overwritten, incomplete, security-rejected, and quality-rejected records as outcomes, not as data to silently remove.

8.4 Metrics

Metric	Definition / reporting recommendation
Bias	Mean signed error by distance and condition.
MAE and RMSE	Central and tail-sensitive error summaries; report both.
Percentiles	Median, 90th, 95th, and, when sample size allows, 99th absolute error.
Availability	Fraction of scheduled procedures yielding a complete accepted result.
Outlier rate	Fraction exceeding a use-case threshold, with threshold declared a priori.
Latency	Connection-to-first-range and procedure-start-to-estimate distributions.
Update rate	Accepted distance outputs per second under the tested schedule.
Energy	Charge or average current per update and at the intended duty cycle.
Security consistency	PBR/RTT disagreement and NADM distribution, including false rejection/acceptance under benign tests.

8.5 Statistical reporting

Report results by distance, environment, orientation, and device pair rather than only as one pooled number. Use confidence intervals obtained at the independent-trial or session level; thousands of correlated updates from one static placement are not thousands of independent experiments. Publish configuration files and raw paired observations where privacy and licensing permit. Distinguish estimator tuning data from evaluation data.

9. Limitations, security, and design trade-offs

9.1 Principal error sources

Source	Effect	Mitigation / reporting
Multipath	Phase and timing can correspond to reflected paths; IFFT may show multiple peaks.	Use channel diversity, antenna diversity, robust estimators, temporal consistency, and environment-stratified results.
Antenna/circuit delay	Frequency- and orientation-dependent bias remains after ideal propagation modeling.	Calibrate the complete device, antenna, enclosure, and operating state; avoid transferring constants between dissimilar devices without validation.
Frequency error	Distorts phase and timing across channels.	Use mode-0 compensation and FAE handling; retain compensation metadata.
Phase wrapping/ambiguity	Multiple distances can explain wrapped phase samples.	Use broad multi-channel data, robust unwrapping/IFFT, and RTT cross-checking.
Motion	The channel changes during a multi-step procedure.	Reduce coherent window, use motion-aware quality checks, and characterize dynamic scenarios separately.
Packet/segment loss	Local and remote records may be incomplete or mismatched.	Use RAS counters/status, on-demand recovery where available, and explicit completeness gates.
Interference	Reduces packet/tone quality and creates uneven channel availability.	Use channel classification, quality filtering, repeated trials, and RF occupancy logging where possible.

9.2 Security model

Channel Sounding adds security mechanisms beyond ordinary connection encryption: randomized CS access addresses and sequences, randomized channel and antenna-path behavior, PBR/RTT diversity, attack detection, and the Normalized Attack Detector Metric (NADM). Some RTT modes include sounding or random sequences, and the LE 2M 2BT PHY was designed to make certain early-commit attacks harder. [1], [2]

These mechanisms are inputs to a secure application, not a complete authorization policy. A digital lock, for example, should require authenticated identity, fresh and complete CS results, an acceptable CS security level, attack metrics below policy thresholds, PBR/RTT consistency, and application-specific timing and distance conditions. RAS characteristics require encryption, but the product must still define bonding, key management, replay resistance, failure handling, and privacy policy. [4], [5]

9.3 Accuracy, latency, power, and security

Design choice	Likely benefit	Likely cost or risk
More PBR channels/steps	More frequency diversity and potentially better multipath discrimination.	Longer procedure, more energy, and greater exposure to motion-induced nonstationarity.
More RTT steps	Stronger independent cross-check and ambiguity/security support.	Additional airtime and often lower fine-ranging precision than PBR.
Shorter procedure interval	Higher update rate and lower tracking latency.	Higher radio/CPU duty cycle and possible coexistence pressure.
Real-time RAS transfer	Lower delivery latency and smaller server storage need.	Weaker recovery semantics and continuous GATT traffic during procedures.
On-demand RAS transfer	Retention, acknowledgement, and optional segment recovery.	Extra control-point round trips, storage, and latency.

Design choice	Likely benefit	Likely cost or risk
Aggressive filtering	Lower transfer and processing cost.	May remove fields needed for later algorithm comparison or quality diagnosis.

9.4 Interoperability caveats

- Both controllers must support Channel Sounding; Bluetooth 6.x branding alone does not guarantee that every optional CS capability or mode is present.
- A standardized radio exchange and RAS transport do not imply a standardized distance output; estimator behavior remains implementation-specific.
- Capabilities must be negotiated. Do not hard-code mode 3, antenna paths, a PHY, or later Core 6.3 enhancements without discovering mutual support.
- Conformance claims must use the applicable Bluetooth qualification path and mandatory RAS errata; running a sample successfully is not itself a product qualification.
- A private GATT characteristic is interoperable only by prior agreement and must not be presented as implementation of RAP/RAS.

10. Conclusions and recommendations

Bluetooth Channel Sounding separates measurement acquisition from distance estimation. The Core standard gives two radios a secure, synchronized way to collect phase and timing observations; HCI delivers those observations to each Host. RAP and RAS then provide the adopted GATT path by which a distance-measurement application obtains the remote device's result. This modularity preserves interoperability while allowing research and product differentiation in calibration, multipath processing, fusion, tracking, latency, and power management.

For the planned two-nRF54L15 system, Device A should initially use the Nordic RAS Initiator/Ranging Requester application and Device B the RAS Reflector/Ranging Responder application. This gives the clean common mapping of Central/client/Initiator/RREQ and Peripheral/server/Reflector/RRSP. Development should first preserve and log raw paired Ranging Data, then compare PBR phase-slope, PBR IFFT, RTT, and fused estimators under controlled distance, orientation, multipath, and motion conditions.

Recommended first milestone Demonstrate complete RAS on-demand transfer with zero missing segments, capture paired local and remote step records for a static line-of-sight distance sweep, and reproduce every printed range offline from the logged raw data. Only then increase update rate or add temporal filtering.

Appendix A. Implementation checklist

- ☐ Pin the exact nRF Connect SDK/toolchain release and archive the generated configuration.
- ☐ Confirm local and remote CS capabilities before choosing PHY, modes, antenna paths, and timing.
- ☐ Establish ACL and CS security; verify encrypted access to every RAS characteristic.
- ☐ Read RAS Features and select a supported transfer mode.
- ☐ Aggregate all HCI Result and Result Continue events without truncation.
- ☐ Preserve Ranging Counter, segment index, completion state, and overwrite events.
- ☐ Pair local and remote subevents before estimation and reject mismatched configurations.
- ☐ Log raw phase/timing data and quality/security metrics before filtering.
- ☐ Calibrate antenna and circuit delays for the complete hardware configuration.
- ☐ Evaluate accuracy, availability, latency, power, and outlier behavior separately.

- ☐ Apply Erratum 26610 and the qualification requirements relevant to the product release.

Appendix B. Acronyms

Acronym	Meaning
ACL	Asynchronous Connection-Oriented logical transport / LE ACL connection
ATT	Attribute Protocol
CS	Channel Sounding
EATT	Enhanced Attribute Protocol bearer
FAE	Frequency Actuation Error
FFO	Fractional Frequency Offset
GAP	Generic Access Profile
GATT	Generic Attribute Profile
HCI	Host Controller Interface
NADM	Normalized Attack Detector Metric
PBR	Phase-Based Ranging
RAP	Ranging Profile
RAS	Ranging Service
RREQ	Ranging Requester
RRSP	Ranging Responder
RTT	Round-Trip Timing

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- [5] **Bluetooth SIG.** [Ranging Profile 1.0](#). Adopted requester/responder profile and procedures.
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